

AI Assistants in Mathematical Research

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ADP Workshop

Agenda

- ① Basic concepts: model, context, tools, cost (10 minutes)
- ② Examples of leading a useful LLM conversation (10 minutes)
- ③ ADP frontend overview (5 minutes)
- ④ Hands-on work in ADP (50 minutes)

What happens when you send a message?

A user sends a request.

The selected **model** receives the current **context**, may use available **tools**, and the request consumes **cost/credits** depending on model choice, input size, files, output length, and tool use.

Concept	Practical meaning
Model	The LLM generating the answer
Context	The message, uploaded files, the current conversation.
Tools	web-search, calculator, python interpreter, text editor, etc.
Cost	Depends on the model and the length of the context, reasoning, and output.

The ChatGPT frontend

The screenshot displays the ChatGPT web interface. On the left is a sidebar with a 'Recents' section containing links like 'Beamer Table Formatting' and 'Workshop ChatGPT Overview'. Below this is a 'Rényi AI Business' section with an 'Invite team members' button. The main chat area shows a message from the user asking to 'Go to: Workspace settings → Billing → Spend controls' and then for the assistant to perform tasks for a user, such as removing a credit cap or verifying the workspace balance. The assistant's response is partially visible, starting with 'If all of the above is correct...'. At the bottom, there is a text input field with the placeholder 'Ask anything', a model selection dropdown currently set to 'Extended', and a microphone icon. A small disclaimer at the very bottom states: 'ChatGPT can make mistakes. OpenAI doesn't use Rényi AI workspace data to train its models.'

The ADP frontend

The screenshot displays the ADP frontend interface in a web browser. The address bar shows the URL `adp.hun-ren.hu/c/ce7d3cb0-18cc-464a-843a-c4f5a1fa696b`. The interface is divided into three main sections:

- Left Sidebar (Chats):** A list of chat sessions. Under "Today", there are four chats: "Erdős-conjecture", "Newton-Leibniz Formula LaTeX Code", "Lovász Local Lemma Explanation", and "Moser Spindle Unit Distance Graph". Under "Previous 7 days", there are ten "New Chat" entries. A "Previous 30 days" section is partially visible at the bottom.
- Main Chat Area:** Displays a conversation with "ChatGPT 5.4 + web". The chat history includes:
 - A message from the user: "considered a disproof rather than just an improved construction, or whether this is actually attributable to 'ChatGPT' specifically versus a different OpenAI reasoning model."
 - A response from "Gyozo Csoka": "How did they found the counterexample?"
 - A response from "ChatGPT 5.4":

They say it was found by an **internal OpenAI general-purpose reasoning model** that was **not specially trained for mathematics** and was **not using a dedicated search scaffold**. According to OpenAI, the model's internal reasoning was **dominated by attempts to construct a counterexample** to the usual upper bound, rather than to prove the conjecture. (openai.com)

At a high level, the proof itself started from a familiar geometric idea and pushed it in a new direction: it used **algebraic number theory**, replacing the Gaussian integers with more general number fields that have richer symmetries and can produce many more unit-distance differences. OpenAI says the argument depends on tools like **infinite class field towers** and **Golod-Shafarevich theory** to show the needed number fields actually exist. (openai.com)
- Input Area:** A text box at the bottom with the prompt "Tell me more about the counterexample" and a microphone icon.

At the bottom of the interface, the text "LibreChat v0.8.6-rc1 - Every AI for Everyone. | Terms of service" is visible.

How to lead a useful conversation

Good conversations are steered.

- ① Give the task, not only the question.
- ② State the goal, definitions, and assumptions.
- ③ Ask for approaches before asking for a full proof.
- ④ Separate exploration from verification.

Common mistakes

Mistake	Better habit
"Can you solve this?"	Give context and what you tried.
Asking for a complete proof too early	Ask first for strategies or obstructions.
Continuing a messy long chat	Summarize and restart cleanly.
Uploading a file and asking vaguely	Ask first for structure, then focused questions.
Using the strongest model for everything	Match model choice to task and cost.
Trusting a polished answer	Verify line by line.
"Is this correct?"	Ask for edge cases or counterexamples.
Assuming references are real	Check references externally.

LLMs are great at translation, you can talk to it in your native language, it understands it just like latex or python. In context-sensitive translation tasks it might choose incorrect words in languages other than English:

- ① bipartite graph: **bipartit gráf** instead of **páros gráf**
- ② expected value: **elvárt érték** instead of **várható érték**
- ③ random variable: **véletlen változó** instead of **valószínűségi változó**
- ④ field: **mező** instead of **test**

Topic	Rule of thumb
Privacy	Do not enter personal data unless clearly allowed.
Confidentiality	Do not upload confidential or third-party unpublished material without permission.
Person-affecting decisions	Do not use the system for decisions about real people.
Responsibility	You remain responsible for proofs, code, text, and claims.
Disclosure	Substantial use may need to be disclosed.